

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q16-0582

Speed 138 kV
August 15, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0582. The Interconnection Customer proposes a 278.8 MW solar photovoltaic facility interconnecting at the Speed 138 kV substation in the Kentucky area. The request was received on February 19, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 248 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$47,963,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Gaga to Barnham 138 kV	Silver Bay Power to Speed 138 kV	127.3	190	100.0
Speed to Gaga 138 kV Line	Silver Bay Power to Speed 138 kV Line	124.8	209	100.0
Speed to Silver Bay Power 138 kV L	Gaga Substation to Speed 138 kV	123.0	212	100.0
N Appleton to Leslie 138 kV	Barnabie Cable Sta to Casino 138 kV	120.2	224	7.2
Leslie to St Joseph 138 kV L	Casino to Barnabie Cable Sta 138 kV L	119.2	221	7.2
Narod to Barnham 138 kV	Silver Bay Power to Speed 138 kV	116.5	191	100.0
Sand Springs to Ojo 138 kV	Sand Springs to Snohomish 138 kV L	109.0	232	6.9
Dismal Riv to Becker 345 kV line	Dillard-Orange to Fishkill Plains 345 kV	103.5	1,250	5.5

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Gaga to Barnham 138 kV	Rebuild line with higher-capacity conductor	\$3,546,000	100% (\$3,546,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Speed to Gaga 138 kV	Rebuild line with higher-capacity conductor	\$5,109,000	100% (\$5,109,000)
Speed to Silver Bay Power 138 kV L	Rebuild line with higher-capacity conductor	\$12,018,000	100% (\$12,018,000)
N Appleton to Lesle 138 kV	Rebuild line with higher-capacity conductor	\$5,899,000	7% (\$408,000)
Leslie to St Joseph 138 kV L	Rebuild line with higher-capacity conductor	\$15,143,000	7% (\$1,047,000)
Narod Substation to Barnham 138 kV	Rebuild line with higher-capacity conductor	\$21,301,000	100% (\$21,301,000)
Sand Springs to Ojo 138 kV	Reconductor line	\$2,657,000	10% (\$273,000)
Becker to Dismal River 345 kV	Reconductor line	\$56,191,000	8% (\$4,261,000)
	Total allocated to project		\$47,963,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.